



BUSINESS ANALYSIS

Lecture 8: Requirements Documentation

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REQUIREMENTS DOCUMENTATION FUNDAMENTALS

- **What is requirements documentation**

“Requirements documentation describes the business solution for a particular project, including the user's expectations and needs, the purpose behind that solution, and any high-level constraints that can affect successful deployment.”

- **Purpose of requirements documentation**

- Provides guideline for stakeholders to decide on project priorities, structure, and design
- Serves as a fundamental contract between the vendor and customer, outlining the project's deliverables
- Sets the standards for determining when you have completed a project

Source: indeed; <https://ca.indeed.com/career-advice/career-development/requirements-documentation>

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REQUIREMENTS DOCUMENTATION FUNDAMENTALS

Persons involved in the creation of a requirements document

- **"The Customer"** is ultimately responsible for determining the requirements. The customers' needs are the origin of the project.
- **The Business Analyst** is responsible for discovering the problem/requirements and determining the solution.
- **The Project Manager** is responsible for delivering the solution to a problem.
- **The Systems Analyst** uses analysis and design to satisfy business requirements using information technology.
- **The Marketing Manager** develops the marketing strategy for the project in line with its requirements.
- **The Product Manager** is responsible for defining the why, when, and what of the product that the development team will build"

Source: modern analyst.com; <https://www.modernanalyst.com/Resources/Articles/tabid/115/ID/5464/9-Types-Of-Requirements-Documents-What-They-Mean-And-Who-Writes-Them.aspx>

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TYPES OF REQUIREMENT DOCUMENTS

Business Requirements Document (BRD)

Describes features the project should provide, which can be interpreted as goals for the development team

- **Who prepares a BRD?**
Project manager or business analyst
- **What it includes**
 - An outline of the requirements of the project
 - Objectives of the project
 - A needs statement detailing why the project is needed and how it will meet those needs
 - Financial statements, demonstrating how the project will be funded and its effect on the company's balance sheet
 - Functional requirements and features
 - A SWOT analysis of the business and how the project fits into it
 - Personnel needs. Who do we want to work on the project?
 - Schedule, timeline and deadlines
 - A cost-benefit analysis

Source: modern analyst.com; <https://www.modernanalyst.com/Resources/Articles/tabid/115/ID/5464/9-Types-Of-Requirements-Documents-What-They-Mean-And-Who-Writes-Them.aspx>



TYPES OF REQUIREMENT DOCUMENTS

Software Requirements Document or Software Requirements Specification (SRS)

An SRS outlines the features and the intended behavior of a system

- **Who prepares an SRS?**

Lead engineer of a project

- **What it includes**

- A product overview
- A summary of the current system
- The proposed methods and procedures
- Design considerations
- Security considerations

Source: modern analyst.com; <https://www.modernanalyst.com/Resources/Articles/tabid/115/ID/5464/9-Types-Of-Requirements-Documents-What-They-Mean-And-Who-Writes-Them.aspx>

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TYPES OF REQUIREMENT DOCUMENTS

User Story

A user story describes a software feature from the perspective of the end user. Example: "As an Author, I want to be able to edit a post, so that I can make necessary changes." The user story follows a structure: As a [persona], I [want to], [so that]. The structure can also be summarized as "Who, What, Why".

Use Case

A use case describes an interaction of a user with a system. It captures the details of a user, their goals, and the required steps to accomplish those goals. Use case is best described as a table.

Difference between User Stories and Use Cases

	USER STORIES	USE CASES
Purpose	Answer who, what and why from a user's perspective	Detailed description of user-system interaction
Level of detail	Brief and concise	In-depth, covering different scenarios
Planning value	Useful for outlining and planning upcoming work	Essential for understanding, verifying, and documenting business requirements
Primary users	Focus on end-users, customers, or stakeholders	Primarily used by development team members

Source: VALA; <https://www.valagroup.com/blog/requirements-engineering-user-stories-vs-use-cases/>

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TYPES OF REQUIREMENT DOCUMENTS

Acceptance Criteria

Acceptance criteria are a range of conditions that a given product must satisfy before the end-user, consumer, or group of stakeholders accepts it

- **When is it written**

It is common practice to write the acceptance criteria while the user story is being discussed, as this usually helps a team with estimation.

- **Three aspects of a good acceptance criterion**

- Functional and non-functional requirements
- Expected performance: Example, 1-3 seconds to query a response. Should always be stated as a goal, and not a solved solution
- Outstanding defects: Defines the acceptable parameters of outstanding open defects

- **Example of acceptance criterion**

The acceptance criteria should be written in a simple to understand language that the customer can decipher, similar to the user story. It can include something like:

- Limit the search by format and type.
- Delineate the search by a date range.
- Limit the search to publisher info like author, title, subject, publisher, and place.
- Restrict the search to specific collections and brands.
- Filter search by availability

Source: CODEFIRST; <https://www.codefirst.co.uk/blog/the-importance-of-acceptance-criteria/>

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TYPES OF REQUIREMENT DOCUMENTS

Requirements Traceability Matrix

A traceability matrix is a document that maps product requirements to the artifacts created during the development and testing process. It shows the relationships and connections between specific requirements, technical specifications, design elements, test plans, and test cases.

How is an RTM developed

An RTM starts by bringing all stakeholders together to define what the product should do. The team discusses customer and regulatory requirements to build a shared understanding of the end goal. RTM involves the following steps:

- Gather Requirements
- Identify Relationships
- Create the Matrix
- Maintain and Update
- Validate Consistency

Source: Chisel blog; <https://chisellabs.com/blog/product-requirements-traceability-matrix/>

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ASSIGNMENT 1: REQUIREMENTS ENGINEERING PROJECT (5 marks)

- Identify a business/company of interest (imaginary or actual)
- Conduct SWOT analysis
- Identify a key requirement (think of it as a project, i.e., it should have a deliverable, a plan, and is time & cost bound)
- For the remaining assignment, refer to the Linear Requirements Engineering Model and the steps involved in mapping the requirement. Then:
 - Prioritize stakeholders through the Power-Interest Grid
 - Identify business requirements, stakeholder requirements and system requirements
 - Identify functional requirements, non-functional requirements and domain requirements
 - Consult the “RE Technique Selection Factors” by Anwar and Razali, and select the RE technique most suitable for your project. Briefly describe it in the context of your project e.g., if you are using interviews, who are you going to interview, why, and how (e.g. structured or unstructured questions)
 - Finally, how are you going to document the requirements? Choose one technique and document your project requirements

Note: Use separate headings for the steps mentioned above

Think scientifically and creatively about your project: How do business analysts resolve issues?

Example 1: adept; <https://adept-plm.com/about-adept/clients/>

Example 2: AISIGHT.ai; <https://www.aisight.ai/>

Source: adept; <https://adept-plm.com/about-adept/clients/>

AISIGHT.ai; <https://www.aisight.ai/>

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ESSENTIAL READINGS

CODEFIRST: “The Importance of Acceptance Criteria”; available at <https://www.codefirst.co.uk/blog/the-importance-of-acceptance-criteria/>

Chiselblog: “How to Create a Product Requirements Traceability Matrix”, available at <https://chisellabs.com/blog/product-requirements-traceability-matrix/>

indeed: “A Guide to Different Requirements Documentation”, available at <https://ca.indeed.com/career-advice/career-development/requirements-documenta>

Modern analyst.com: “9 Types Of Requirements Documents: What They Mean And Who Writes Them”; available at <https://www.modernanalyst.com/Resources/Articles/tabid/115/ID/5464/9-Types-Of-Requirements-Documents-What-They-Mean-And-Who-Writes-Them.aspx>

VALA: “Requirements Engineering: User Stories vs. Use Cases”; available at <https://www.valagroup.com/blog/requirements-engineering-user-stories-vs-use-cases/>

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REFERENCE BOOKS OPTIONS

- **BUSINESS ANALYSIS BODY OF KNOWLEDGE (BABOK)**; IIBA Bookstore: <https://www.iiba.org/career-resources/a-business-analysis-professionals-foundation-for-success/babok/>
- **BUSINESS ANALYSIS (Fourth Edition)** by Debra Paul (Author), James Cadle (Author), Malcolm Eva (Author), Craig Rollason (Author), Jonathan Hunsley (Author)
- **BUSINESS ANALYSIS TECHNIQUES: 123 ESSENTIAL TOOLS FOR SUCCESS (3RD Edition)** by James Cadle (Author), Debra Paul (Author), Jonathan Hunsley (Author), Adrian Reed (Author)